

URL Shortener - Requirement Analysis

Functional Requirements

1. Generate short URL
2. Redirect short URL to original URL
3. Store URL mapping
4. Handle invalid URL
5. URL expiry
6. Analytics

Non-Functional Requirements

1. Low Latency (<100 ms)
2. High Scalability
3. Reliability
4. Durability
5. Security
6. Consistency

Estimation

Assumptions

1. 100 Million new URLs per day
2. Read : Write = 100 : 1
3. One URL record = 300 bytes
4. One server can handles = 1000 requests per sec

1. No. of Servers

Write Traffic

100 Million / 24*60*60

≈ 1,157 writes/sec

≈ 1,200 writes/sec

Total Traffic

120,000 + 1,200

≈ 121,200 requests/sec

Assume 1 server = 1,000 requests/sec

121,200 / 1,000

≈ 122 servers

≈ 150 Application Servers

Read Traffic

1,200 × 100

≈ 120,000 reads/sec

2. Data Storage

One Record = 300 Bytes

Per Day

100 Million × 300 Bytes

= 30 GB/day

Per Year

30 × 365

≈ 11 TB/year

For 5 Years

For 5 Years

$11 \times 5 \approx 55 \text{ TB}$

3. RAM Estimation (Cache)

Assume we cache **10 Million popular URLs**

Memory per cached record $\approx$ **300 Bytes**

$10 \text{ Million} \times 300 \text{ Bytes}$

$\approx 3 \text{ GB}$